

Response Letter with Annotations

Richardson, R. A. K., Hong, S. S., Byrne, J. A., Stoeger, T., & Amaral, L. A. N. (2025). The entities enabling scientific fraud at scale are large, resilient, and growing rapidly. Proceedings of the National Academy of Sciences, 122(32). <https://doi.org/10.1073/pnas.2420092122>

Annotations

1. Purposes

a. Motivation (one-sentence summary)

The authors conducted the study because they believe ethics and the pursuit of knowledge for the greater good are diverging due to misconduct rising among scientists and quality of papers decreasing overall – and they believe this is a threat to the validity of scientific research.

b. Specific Rationale (several bullet points)

The authors were concerned about different types of scientific fraud, which they believe are enabled by the wrong kinds of incentives in the academy:

- Scholarly defection, or a “failure to make genuine contributions to the production of knowledge or training of an expert workforce while still benefitting from a funding contract”
- Falsifying research findings, where a systematic analysis found over 20,000 articles with “inappropriately duplicated” images with at least half suggesting the images were deliberately manipulated
- Paper mills, where people churn out low quality, mass produced and fabricated research articles (fueled by the “publish or perish” problem in academia)

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
- Identify specific relationships among variables

The authors used regression modeling, descriptive statistics, and network analysis methods to assess how individuals and organizations that produce fraudulent articles and circumvent ethical and peer-review processes may be connected to one another. They overall hypothesized that fraud is widespread, increasing, and paper mills are increasing their capacity with some connections among editors at different journals. They also hypothesized that retractions, PubPeer comments, and deindexing would increase based on paper mill association.

2. Method

a. Research Design

The authors identified networks of individuals and other entities that produce scientific fraud “at scale” through using indexed journal data from Elsevier’s Scopus, PubMed/MEDLINE, and OpenAlex (which includes open source, standardized data from Microsoft Academic Graph, Crossref, ORCID, Unpaywall, and “other institutional repositories”).

b. Sample (representative sample?)

The authors sampling method is a bit more complicated than representative or not, because there are multiple samples and data sources involved. The indexed journal data from databases were cross checked with other lists (in the Measures section below).

c. Measures

The study evaluated author misconduct across a few areas:

- Analyzing editors of potentially fraudulent articles
- Retractions of articles and PubPeer comments from Retraction Watch
- Falsifying results or duplicated figures provided by PubPeer. These were used in order to create a network model in which each node represents an article, and edges “indicate sharing of an image”.
- Deindexing of journals cross matched from the Academic Research and Development Association (ARDA)’s list of journals “in which they can guarantee publication” (because they’re a now known, large scale paper mill) and constructed a dataset following any deindexation of journals over time shared between the two using the Way Back Machine.
- Suspected paper mill corpus compiled by experts over time that look for “unexpected similarities” and shared content across journals and articles.

d. Analysis

The authors conducted their analyses in several parts:

- To assess editor influence on the “false discovery rate” (FDR), they used Poisson binomial regression models to predict if an editor accepted more “eventually retracted” publications significantly more often than by random chance, and to predict if there are authors who are assigned to these editors more than expected by chance.
- They also investigated articles with turnaround times “30 days or fewer between submission and acceptance” (highly associated with paper mill strategy)
- They then conducted a network analysis of the publishing relationship among the editors (who had their names associated with articles they edited and published) and authors to analyze their connectedness.
- They assessed the pervasiveness of image duplication through creating a network of flagged articles and assessed how concentrated these publications were over time and “modules” which were identified by stochastic block modeling
- They assessed journal targeting and journal hopping by paper mills such as ARDA by cross referencing their “guaranteed publication” list with journal indexes by Scopus and MEDLINE
- Lastly, they looked at general trends of retracted articles and PubPeer comment rates over time.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

The study's findings were broken down into four core sections:

- The authors found with regard to “editorial handling of problematic publications” that, among publications who had handling editors listed as part of the article, these articles “were retracted significantly more frequently than one would expect by chance” and rejected the null hypothesis. The findings were similar when adjusting for authors who directed articles to these editors more frequently.
 - Their network model showed a densely connected network of editors of which over half of this group’s articles had been retracted. They found that 84 IEEE conference proceedings hosted retracted articles “anomalously frequently”.
- The study also found with regard to “coordinated production of fraudulent science” using networks of image duplication that many publications are extremely concentrated in a small amount of time, and by publisher. This supported their hypothesis that paper mills cooperate with individuals at target journals who may impact decision-making and therefore can guarantee publications as advertised.
- The study found with regard to “evidence of journal targeting and journal hopping” that slightly over half of paper mill journal targets were indexed by Scopus, slightly over a quarter were indexed by WoS, and ~15 percent were indexed by both. MEDLINE had the least number of these paper mill target journals at 1.1%.
 - The authors noted that ARDA had expanded their scope of journals throughout their study from 14 journals in Jan 2018 to 86 journals in March 2024.
 - They also found the likelihood of a journal listed as a guaranteed publication by ARDA being deindexed exceeded their baseline rate at 33.3%, and the pattern of deindexing seems to follow ARDA’s listing of journals on their site.
- Lastly, the study showed “differential prevalence” of fraud within disciplinary sub-fields, where they found ARDA-related articles were seemingly off-topic and out of scope for the journals in which they were published and submitted to.
 - Retracted articles and PubPeer comments on articles have increased exponentially over the last 30 years.

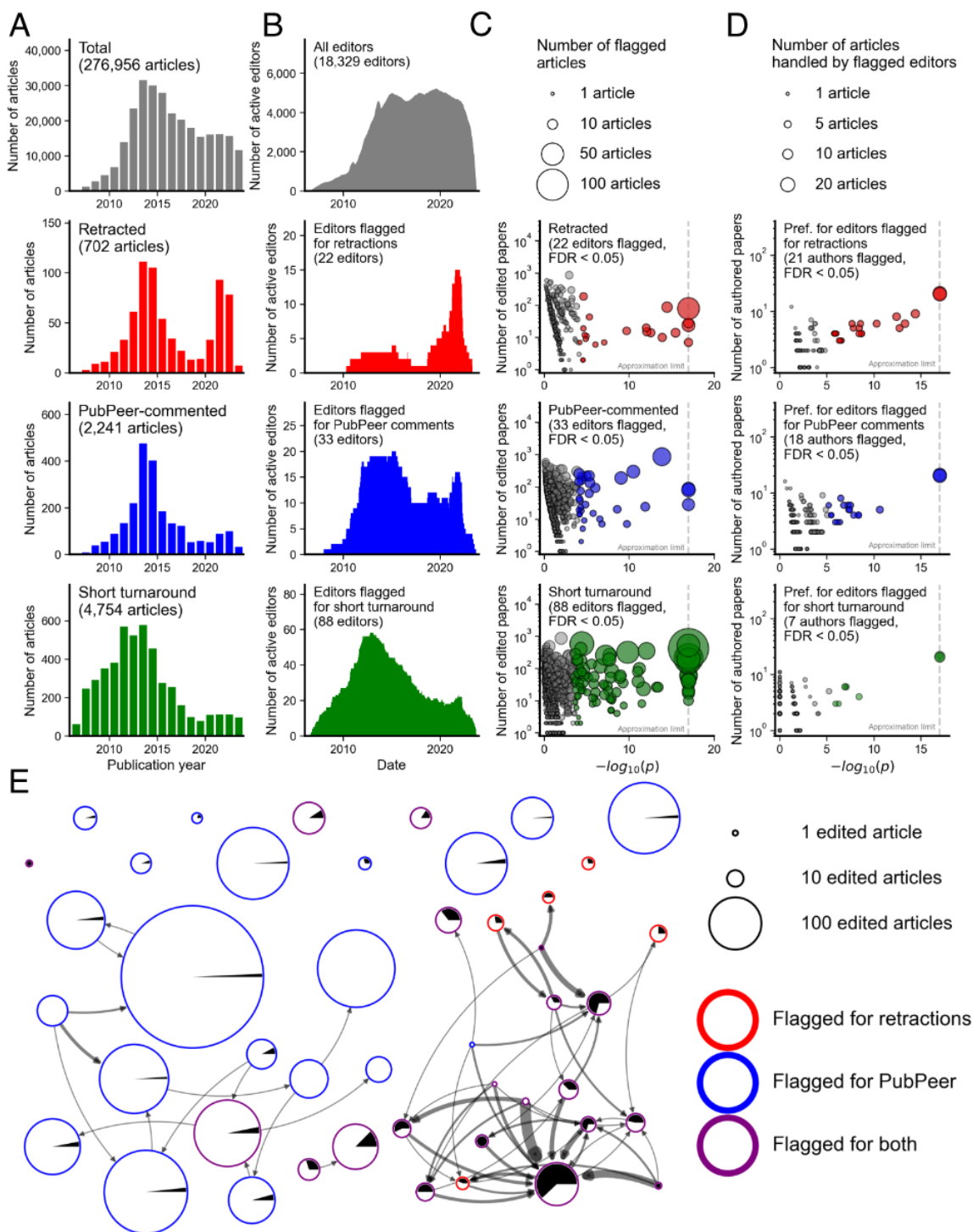


Fig. 1. Evidence of coordination between editors and authors for the publication of fraudulent science in *PLOS ONE*. (A) Number of articles published annually in *PLOS ONE* since its inception (*Top row*) and the annual number of articles retracted (second row), receiving PubPeer comments (third row), or having been accepted 30 or fewer days after submission (*Bottom row*). (B) Number of “active” editors and their tenures at *PLOS ONE* (*Materials and Methods*). Flagged editors tended to have overlapping tenures. (C) Detection of editors accepting an anomalous number of articles which were later retracted (second row), later commented in PubPeer (third row), or accepted after a short turnaround (*Bottom row*). We show each editor as a circle with an area proportional to the number of articles they accepted. At a false discovery rate (FDR) of 0.05, we identify 22 editors who accepted an anomalous number of articles later retracted, 33 editors who accepted an anomalous number of articles later receiving PubPeer comments, and 88 editors who accepted an anomalous number of articles after a short turnaround. (D) Detection of authors whose accepted articles are assigned at an anomalous rate to flagged editors for retractions (second row), PubPeer comments (third row), and short turnaround (*Bottom row*). At an FDR of 0.05, we flagged 21 authors for anomalous assignment of their accepted articles to editors flagged for retractions, 18 authors for anomalous assignment of their accepted articles to editors flagged for PubPeer comments, and 7 authors for anomalous assignment of their accepted articles to editors flagged for short turnaround. (E) Publication interactions among individuals flagged for retractions and/or PubPeer comments. We show every individual as a circle whose size is proportional to the number of accepted articles that they handled as editor (range of 1 to 852 articles). The black wedge inside the circle indicates the fraction of that editor’s accepted articles which were later retracted. Individuals flagged for retractions alone have a red boundary, individuals flagged for PubPeer comments alone have a blue boundary, and individuals flagged for both have a purple boundary. The individuals are connected by arrows that point from author to editor, with the width of each arrow proportional to the total number of articles for which that editor served for that author (range of 1 to 10 articles). Flagged individuals frequently handled each other’s submissions to *PLOS ONE*, especially within a group of editors serving from 2020 to 2023 (cluster on the *Lower Right*).

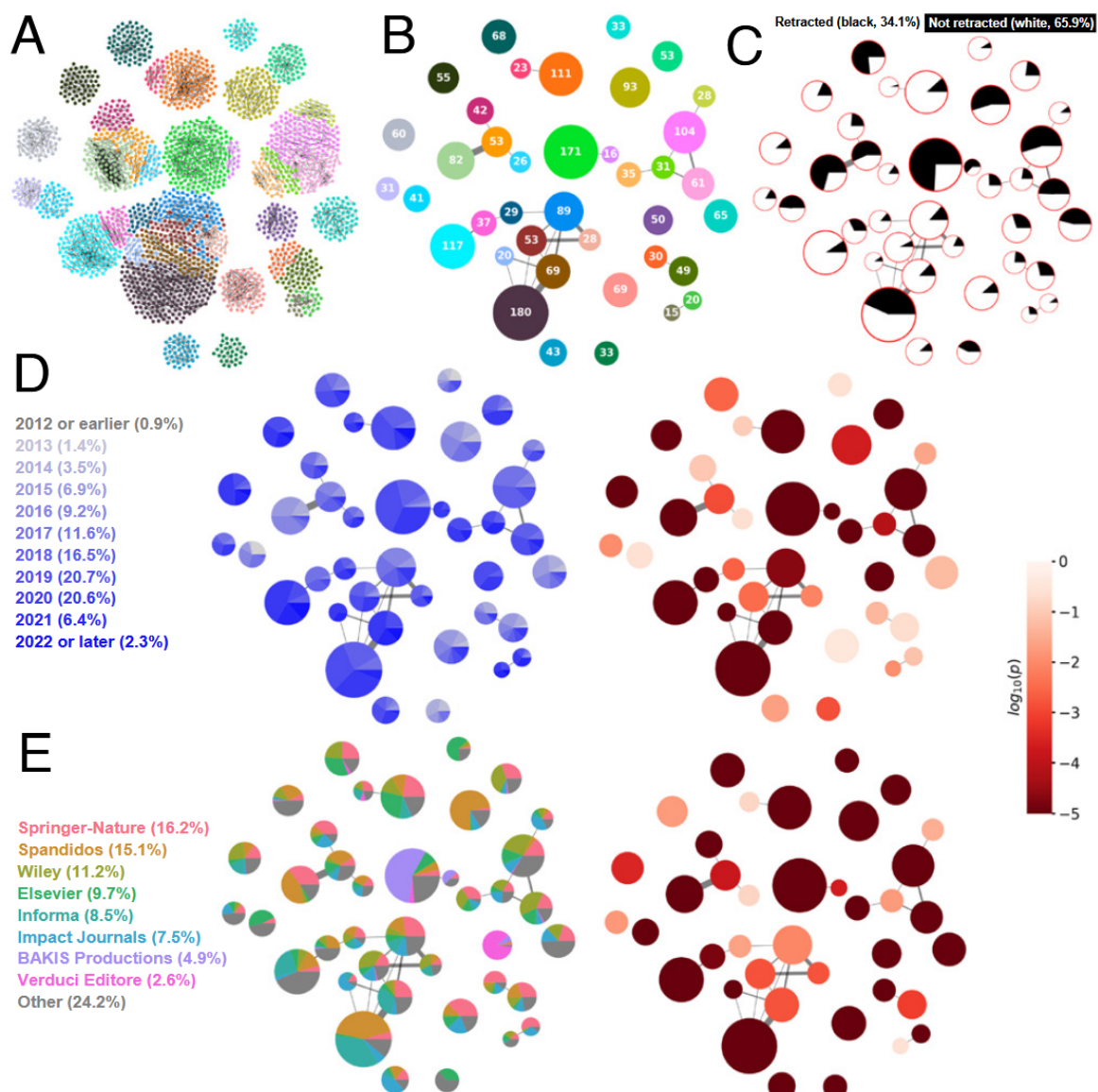


Fig. 2. Evidence of coordination in the production of fraudulent science and the targeting of specific journals for its publication. (A) We build a network of articles with comments in PubPeer concerning image duplication. We identify 2,213 articles (nodes) that are connected through 4,188 edges indicative of image duplication across a pair of articles. We identify 20 connected components within this network. (B) Cartographic representation of the network. Each different colored circle represents a module identified using stochastic block modeling (73, 74). The number inside a circle indicates the number of articles in the module. The lines connecting circles indicate that the two modules are part of the same connected component and their width denotes the number of intermodule edges. (C) Fraction of retracted articles for each module. The black wedge denotes the fraction of retracted articles in the module. (D) Distribution of years of publication for articles in each module (*Left*) and p -value for the null hypothesis that years are distributed randomly across modules (*Right*). Color indicates year of publication (*Left*) or $\log_{10}$ of p (*Right*). (E) Distribution of the publishers of the articles in each module (*Left*) and p -value for the null hypothesis that publishers are distributed randomly across modules (*Right*). Color indicates publisher (*Left*) or $\log_{10}$ of p (*Right*). Dark saturated red indicates extremely low probabilities ($P < 10^{-5}$), suggesting rejection of the null hypothesis. It is visually apparent for both cases that the null hypothesis is rejected for all modules for which the analysis has sufficient power.

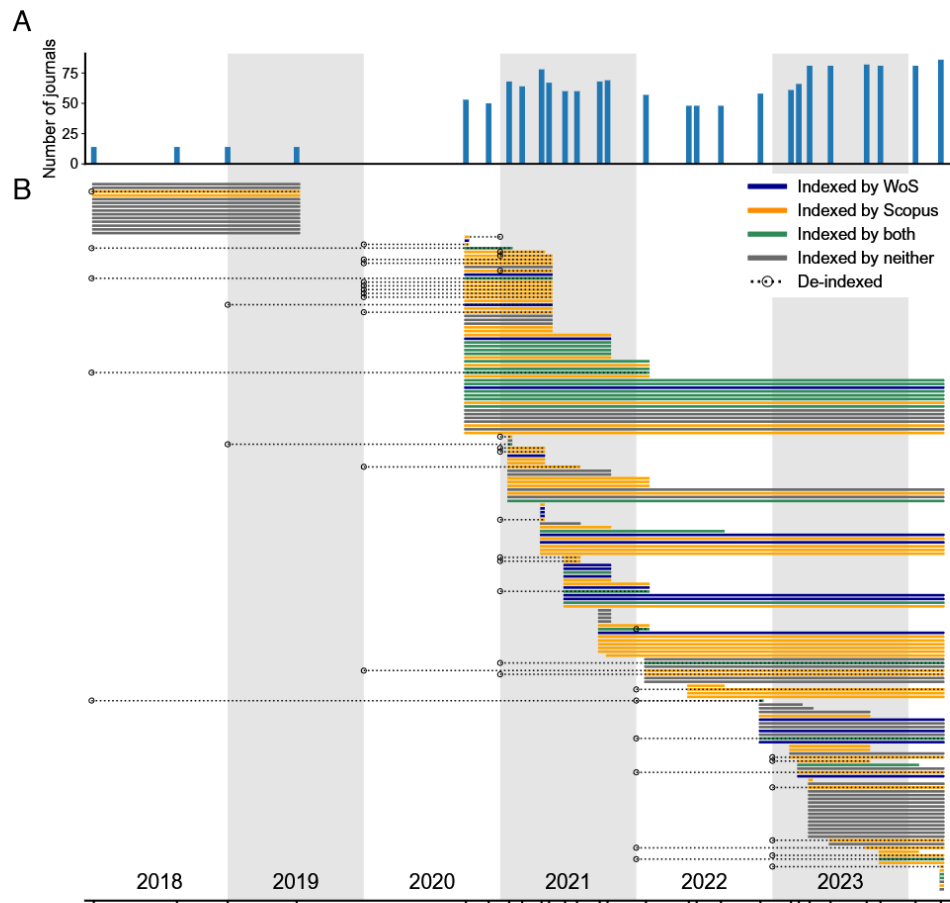


Fig. 3. The business ARDA maintains an evolving portfolio of targeted journals. (A) Number of journals listed on ARDA's website over time. We identify 188 unique journals listed by ARDA. Since 2021, ARDA has maintained a portfolio of 70+ journals in which they guarantee article publication. (B) Estimated time span over which individual journals were listed by ARDA. We represent each journal as a horizontal line and use color to indicate source of journal indexing (WoS, Scopus, or both). We start a bar on the earliest snapshot date we find it listed on ARDA's website and end it on the final snapshot date. We show year of deindexing by an empty circle at the same level as the journal bar, placing it at the start of the year (due to a lack more detailed information on exact timing), followed by a dotted line from this point to the last time the journal was listed on ARDA's website. Note that deindexing can be retroactive (e.g. Scopus could, in 2020, deindex a journal with an effective date in 2018).

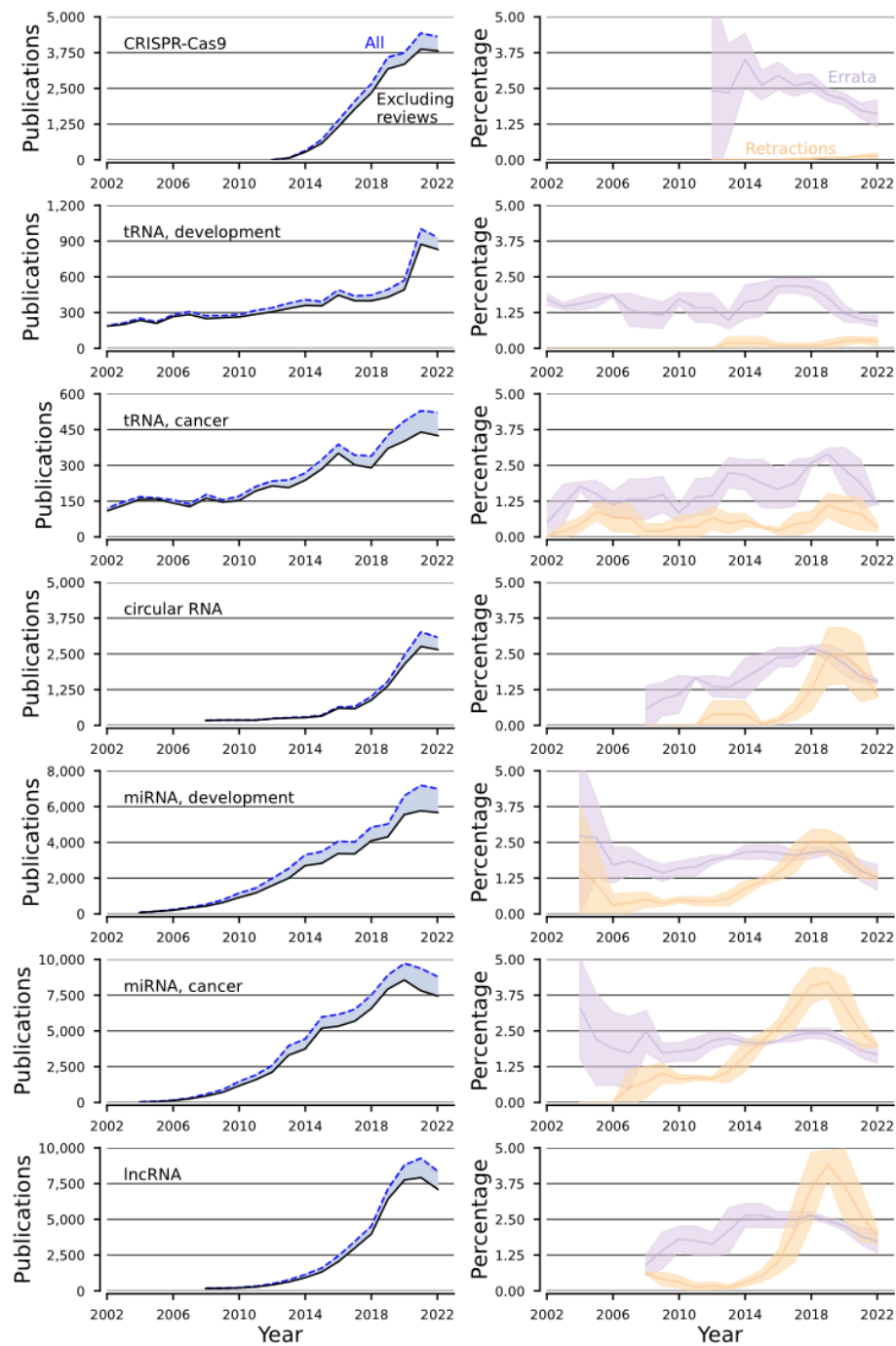


Fig. 4. Closely related subfields of biomedical research display similar rates of errata, but different rates of retractions. We investigated rates of retraction and errata for six subfields within RNA biology, focusing mostly on subfields that had their onset since 2000 and have experienced a dramatic increase in the number of annual publications (*Left column*). The dotted lines show all publications, including reviews, while the full lines exclude reviews from the count. While annual number of publications or their growth rate can differ significantly, all subfields have similar rates of errata. In contrast, the rate of retractions shows remarkable differences across subfields (*Right column*). Most rates decrease after 2019, presumably due to time censoring (i.e., not all retractions or errata may have yet occurred or been reported). Studies concerning “lncRNAs” and “miRNAs and cancer” show retraction rates which peak at around 4%, higher than even the rate at which errata are written. These rates are unlikely to be compatible with good-faith mistakes and point toward concerted attempts at scientific fraud.

4. Discussion

a. Implications

The authors overall concluded that scientific fraud is indeed widespread and growing faster than the base of legitimate scientific research. They attribute this to the competitive environment for funding and tenure that is disproportionate with the resources afforded to all research as an enterprise. This relationship creates misalignment with knowledge for the greater public good. Essentially, when there is a demand for publications regardless of quality or substantive content, there will be people who capitalize on the lack of opportunity, training, and resources for early researchers when conducting research is a high-risk, high reward activity – but failure to produce “exciting” results that can make or break one’s career or consideration for tenure promotion can be devastating.

b. Limitations

The authors make very clear that their data is not necessarily representative and is not comprehensive -- they were only able to assess articles where scientific fraud was already known, or highly probable. They state their estimates are likely on the conservative side, as these are the examples that were caught. Many experts on scientific fraud believe that most paper mill publications have gone undetected. They also assert their findings aren’t inherently generalizable since they focused on some subfields over others. Lastly, they state that fraud of a similar nature may have existed for a long time, and that only now are researchers finding ways to detect these practices systematically.

c. Future Directions

The authors cite how important journal indexing was to their study, and that these repositories and platforms like PubPeer are extremely important to holding other scientists and publications accountable to publishing well designed research. They acknowledge the difficult process it is to change an institution’s culture and incentives, especially one as rigid as the academy. They assert accountability measures need to be more rigorous and penalizing – or the field of science as a whole will continue to suffer threats from other institutions and the general public whose trust in science is slowly dwindling.

Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

This study is fantastic, but its results are concerning -- and does raise alarms about how we need to change this institution's incentives. Because, as things are, "publish or perish" incentives aren't in line with the desired outcomes of scientific research. The authors contended with a lot of different data sources that weren't meant to work together out of the box, and they seemingly had the cooperation of several organizations in conducting this important investigation into scientific fraud and paper mills like ARDA -- whose name makes them seem more legitimate in function than they are in practice.

I think the weaknesses with the study come at the same time as its strengths, and at the restriction of the word count and publishing requirements imposed by PNAS (which I am unfamiliar with the details, but I know it's a very particular journal meant to be concise yet high profile). I think the paper could have benefited from a higher word count to better explain their materials and methods section in greater detail. I'm unsure if I could assemble a team to replicate this study from their description. Some of the connections between the materials used and the areas of concern related to scientific fraud were present in the introduction -- where the authors raise concerns about paper mills, editor influence, and the ways researchers can be sanctioned for misconduct. Considering their visualizations since that is the topic for our seminar this week, I found a lot of the visualizations to be too small -- especially the point data in Figure 1C-D, and the network maps in Figure 2. I think those findings are super important but it's difficult to make out the details of them. They also were not adequately described since the visualizations are simple in nature. Either have a simple visual with some explanation, or have the visual that speaks a thousand words.

I am glad they acknowledged that their sample wasn't representative and only included publications that were already reported. They did a great deal of work to demonstrate these associations among publications and journals that organizations had already flagged. I am left curious as to how the authors and/or other experts in this area would approach detection of concerning articles on a broader scale, and if they think these methods could be built upon as a means of oversight to keep journals and academics honest about their work to a higher degree.